

ME315 - Lab07

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1

Baixe o arquivo uwmadison.sqlite3. Conecte-se a ele usando o pacote RSQLite, armazenando a conexão em uma variável conn.

```
library(DBI)
library(RSQLite)

# Conecta/Cria o arquivo
conn <- dbConnect(RSQLite::SQLite(), "database.sqlite3")
```

2

Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (subjects.abbreviation='STAT');

```
SELECT DISTINCT instructors.name FROM subjects
INNER JOIN subject_memberships ON subject_memberships.subject_code=subjects.code
INNER JOIN sections ON sections.course_offering_uuid=subject_memberships.course_offering_uuid
INNER JOIN teachings ON teachings.section_uuid=sections.uuid
INNER JOIN instructors ON instructors.id=teachings.instructor_id
WHERE subjects.abbreviation='STAT'
ORDER BY instructors.name
```

Table 1: Displaying records 1 - 10

name
ABIGAIL BENZINE
ADIN-CRISTIAN ANDREI
AKICHIKA OZEKI

name
ALAN HUANG
ALBERTO DEL PIA
ALBRECHT KLEMM
ALEXANDER COVINGTON
ALEXANDER FISH
ALEXANDER KISELEV
ALYSSA DIGILIO

```
SELECT COUNT(DISTINCT instructors.name) AS QUANTOS_PROFESSORES FROM subjects
INNER JOIN subject_memberships ON subject_memberships.subject_code=subjects.code
INNER JOIN sections ON sections.course_offering_uuid=subject_memberships.course_offering_uuid
INNER JOIN teachings ON teachings.section_uuid=sections.uuid
INNER JOIN instructors ON instructors.id=teachings.instructor_id
WHERE subjects.abbreviation='STAT'
ORDER BY instructors.name
```

Table 2: 1 records

QUANTOS_PROFESSORES
377

3

O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

- Quem é o professor mais difícil?
- Quem é o professor mais fácil?
- Qual é a disciplina mais difícil?
- Qual é a disciplina mais fácil?

Professor mais difícil:

```
SELECT instructors.name AS PROFS, SUM(
grade_distributions.a_count*4.0+
grade_distributions.ab_count*3.5+
grade_distributions.b_count*3.0+
```

```

grade_distributions.bc_count*2.5+
grade_distributions.c_count*2.0+
grade_distributions.d_count*1.0+
grade_distributions.f_count*0.0
)*1.0/SUM(grade_distributions.a_count+
grade_distributions.ab_count+
grade_distributions.b_count+
grade_distributions.bc_count+
grade_distributions.c_count+
grade_distributions.d_count+
grade_distributions.f_count) AS NOTA_MEDIA
FROM grade_distributions
INNER JOIN subject_memberships ON subject_memberships.course_offering_uuid= grade_distributi
INNER JOIN sections ON sections.course_offering_uuid=subject_memberships.course_offering_uui
INNER JOIN subjects ON subjects.code= subject_memberships.subject_code
INNER JOIN teachings ON teachings.section_uuid=sections.uuid
INNER JOIN instructors ON instructors.id=teachings.instructor_id
WHERE subjects.abbreviation='STAT'
GROUP BY instructors.name
HAVING NOTA_MEDIA IS NOT NULL
ORDER BY NOTA_MEDIA
LIMIT 1

```

Table 3: 1 records

PROFS	NOTA_MEDIA
JAMES D KUELBS	2.607692

Professor mais fácil:

```

SELECT instructors.name AS PROFS, SUM(
grade_distributions.a_count*4.0+
grade_distributions.ab_count*3.5+
grade_distributions.b_count*3.0+
grade_distributions.bc_count*2.5+
grade_distributions.c_count*2.0+
grade_distributions.d_count*1.0+
grade_distributions.f_count*0.0
)*1.0/SUM(grade_distributions.a_count+
grade_distributions.ab_count+
grade_distributions.b_count+

```

```

grade_distributions.bc_count+
grade_distributions.c_count+
grade_distributions.d_count+
grade_distributions.f_count) AS NOTA_MEDIA
FROM grade_distributions
INNER JOIN subject_memberships ON subject_memberships.course_offering_uuid= grade_distributi
INNER JOIN sections ON sections.course_offering_uuid=subject_memberships.course_offering_uui
INNER JOIN subjects ON subjects.code= subject_memberships.subject_code
INNER JOIN teachings ON teachings.section_uuid=sections.uuid
INNER JOIN instructors ON instructors.id=teachings.instructor_id
WHERE subjects.abbreviation='STAT'
GROUP BY instructors.name
HAVING NOTA_MEDIA IS NOT NULL
ORDER BY NOTA_MEDIA DESC
LIMIT 1

```

Table 4: 1 records

PROFS	NOTA_MEDIA
YAJUAN SI	4

Disciplina mais difícil:

```

SELECT courses.name AS DISCIPLINAS, SUM(
grade_distributions.a_count*4.0+
grade_distributions.ab_count*3.5+
grade_distributions.b_count*3.0+
grade_distributions.bc_count*2.5+
grade_distributions.c_count*2.0+
grade_distributions.d_count*1.0+
grade_distributions.f_count*0.0
)*1.0/SUM(grade_distributions.a_count+
grade_distributions.ab_count+
grade_distributions.b_count+
grade_distributions.bc_count+
grade_distributions.c_count+
grade_distributions.d_count+
grade_distributions.f_count) AS NOTA_MEDIA
FROM grade_distributions
INNER JOIN subject_memberships ON subject_memberships.course_offering_uuid= grade_distributi
INNER JOIN course_offerings ON course_offerings.uuid=subject_memberships.course_offering_uui

```

```

INNER JOIN subjects ON subjects.code=subject_memberships.subject_code
INNER JOIN courses ON courses.uuid=course_offerings.course_uuid
WHERE subjects.abbreviation='STAT'
GROUP BY courses.name
HAVING NOTA_MEDIA IS NOT NULL
ORDER BY NOTA_MEDIA
LIMIT 1

```

Table 5: 1 records

DISCIPLINAS	NOTA_MEDIA
Introduction to the Theory of Probability	2.886374

Disciplina mais fácil:

```

SELECT courses.name AS DISCIPLINAS, SUM(
grade_distributions.a_count*4.0+
grade_distributions.ab_count*3.5+
grade_distributions.b_count*3.0+
grade_distributions.bc_count*2.5+
grade_distributions.c_count*2.0+
grade_distributions.d_count*1.0+
grade_distributions.f_count*0.0
)*1.0/SUM(grade_distributions.a_count+
grade_distributions.ab_count+
grade_distributions.b_count+
grade_distributions.bc_count+
grade_distributions.c_count+
grade_distributions.d_count+
grade_distributions.f_count) AS NOTA_MEDIA
FROM grade_distributions
INNER JOIN subject_memberships ON subject_memberships.course_offering_uuid= grade_distributions.course_offering_uuid
INNER JOIN course_offerings ON course_offerings.uuid=subject_memberships.course_offering_uuid
INNER JOIN subjects ON subjects.code=subject_memberships.subject_code
INNER JOIN courses ON courses.uuid=course_offerings.course_uuid
WHERE subjects.abbreviation='STAT'
GROUP BY courses.name
HAVING NOTA_MEDIA IS NOT NULL
ORDER BY NOTA_MEDIA DESC
LIMIT 1

```

Table 6: 1 records

DISCIPLINAS	NOTA_MEDIA
Sample Survey Theory and Method	4

4

Desconecte do banco de dados.

```
dbDisconnect(conn)
```

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-23 23:19:50 -03"
```